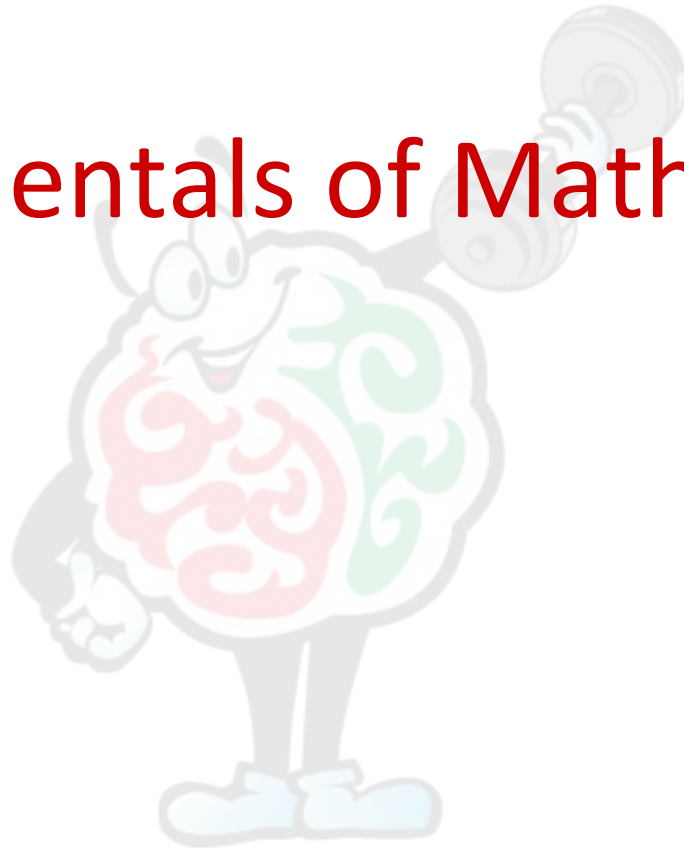


Fundamentals of Mathematics



Arithmetic

- Sum = addend + addend (or terms)
- Difference = minuend – subtrahend
- Product = multiplicand × multiplier (or factors)
- Quotient = dividend ÷ divisor
- Take note! $\frac{a}{0}$ is undefined

Arithmetic

- Addition

- Same sign: add the numbers, copy the sign
- Opposite signs: subtract the numbers, copy sign of larger number

- Subtraction

- Change the operation into addition, then change the sign of the subtrahend

- Multiplication/division

- Count the number of negative signs; when even = positive; when odd = negative

Properties of arithmetic

- Commutative (+,×) $a + b = b + a$
- Associative (+,×)
- Distributive (× over + and ÷ over +)
- Identity element (0 for + and 1 for ×)
- Inverse elements
 - Additive inverses have the sum 0
 - Multiplicative inverses have the product 1
- Multiplicative property of zero

Properties of equality

- Reflexive

$$a = a$$

- Symmetric

$$a = b; \text{ then } b = a$$

- Transitive

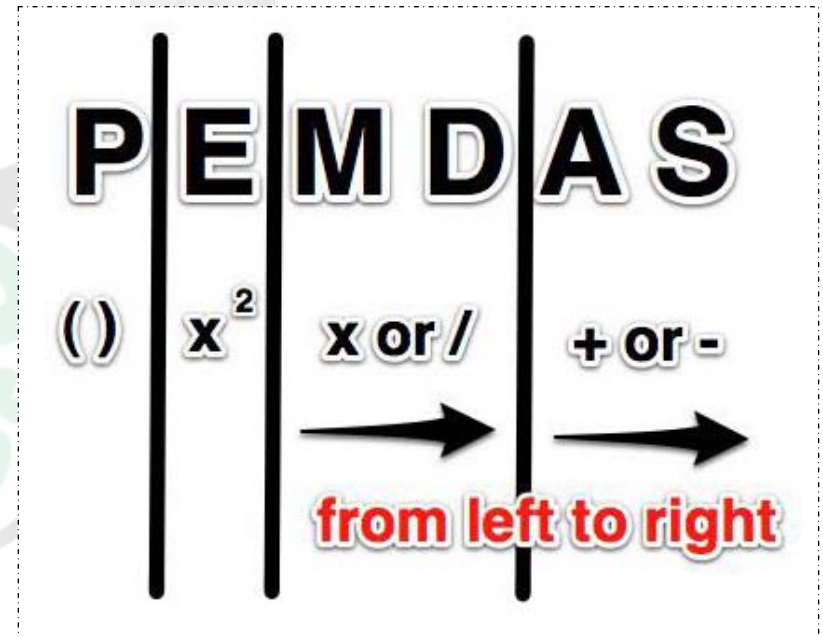
$$a = b \text{ and } b = c, \text{ then } a = c$$

- Substitution

If $a = b$, then any expression involving a retains the same value if a is replaced by b .

Order of operations: PEMDAS

- Parentheses or grouping symbols
- Exponents
- Multiplication and division
(whichever comes first)
- Addition and subtraction
(whichever comes first)



<http://www.mahalo.com/pemdas/>

PRACTICE

$$[5 + (33 - 8)] \div 6 + 8(3 + 4) - 21$$

$$[3 + (6 \times 2^2 + 9 \div 3)] \div 6$$

PRACTICE

$$[(10 \times 2 + 8) \div 14]^2$$

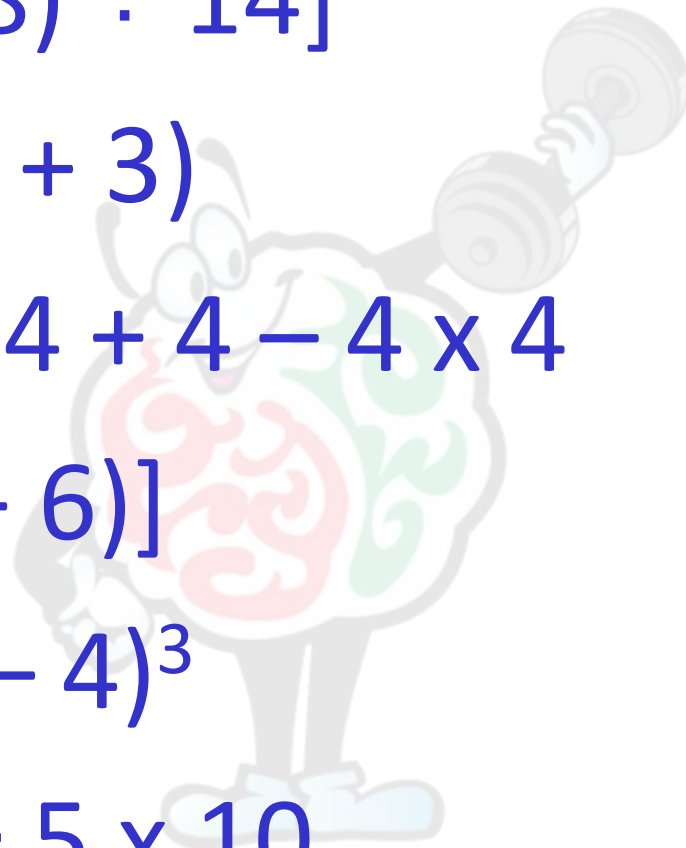
$$7 + (6 \times 5^2 + 3)$$

$$4 \times 4 + 4 \times 4 + 4 - 4 \times 4$$

$$2[13 - (1 + 6)]$$

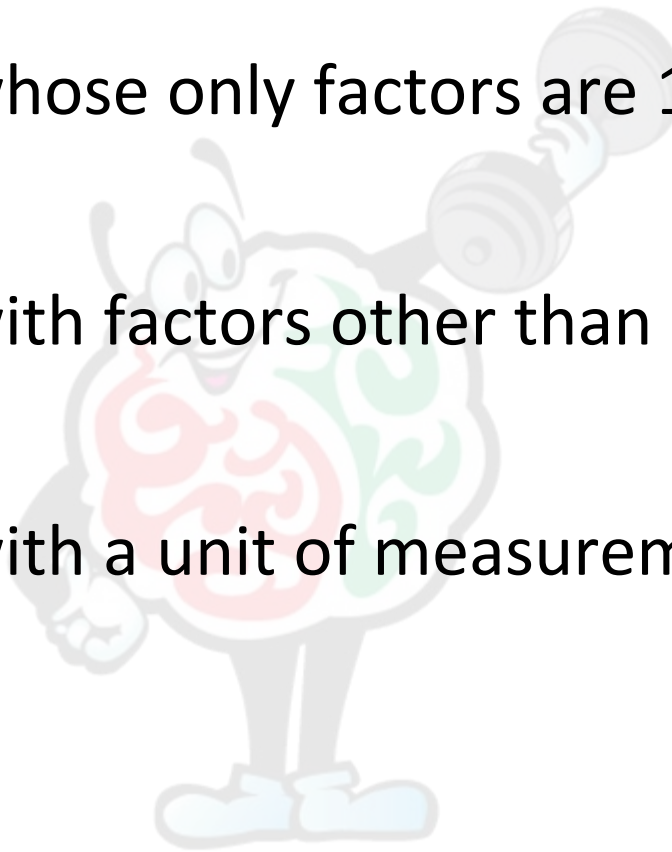
$$100 - 4(7 - 4)^3$$

$$100 - 50 \div 5 \times 10$$



Prime, Composite or Denominate?

- Prime
 - Numbers whose only factors are 1 and itself
- Composite
 - Numbers with factors other than itself and 1
- Denominate
 - Numbers with a unit of measurement



PRACTICE

What are the factors and prime factors of:

35

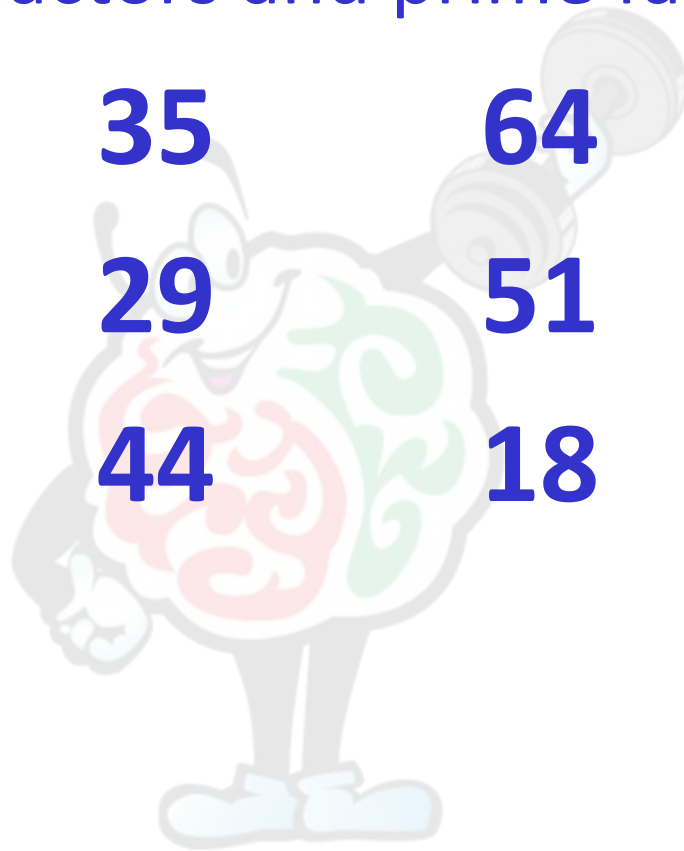
64

29

51

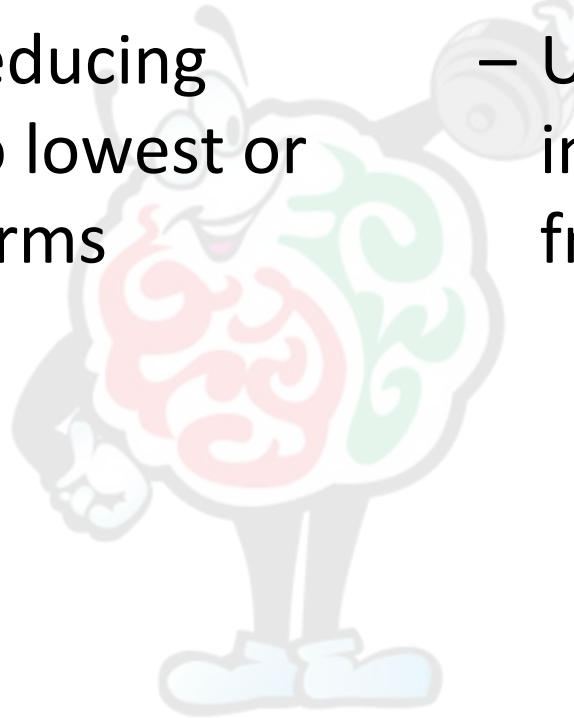
44

18



GCF and LCM

- Greatest common factor
 - Useful in reducing fractions to lowest or simplest terms
- Least common multiple
 - Useful in operations involving dissimilar fractions



PRACTICE

What is the GCF of these pairs?

35 and 51

24 and 108

64 and 18

16 and 48

What is the LCM of these pairs?

24 and 108

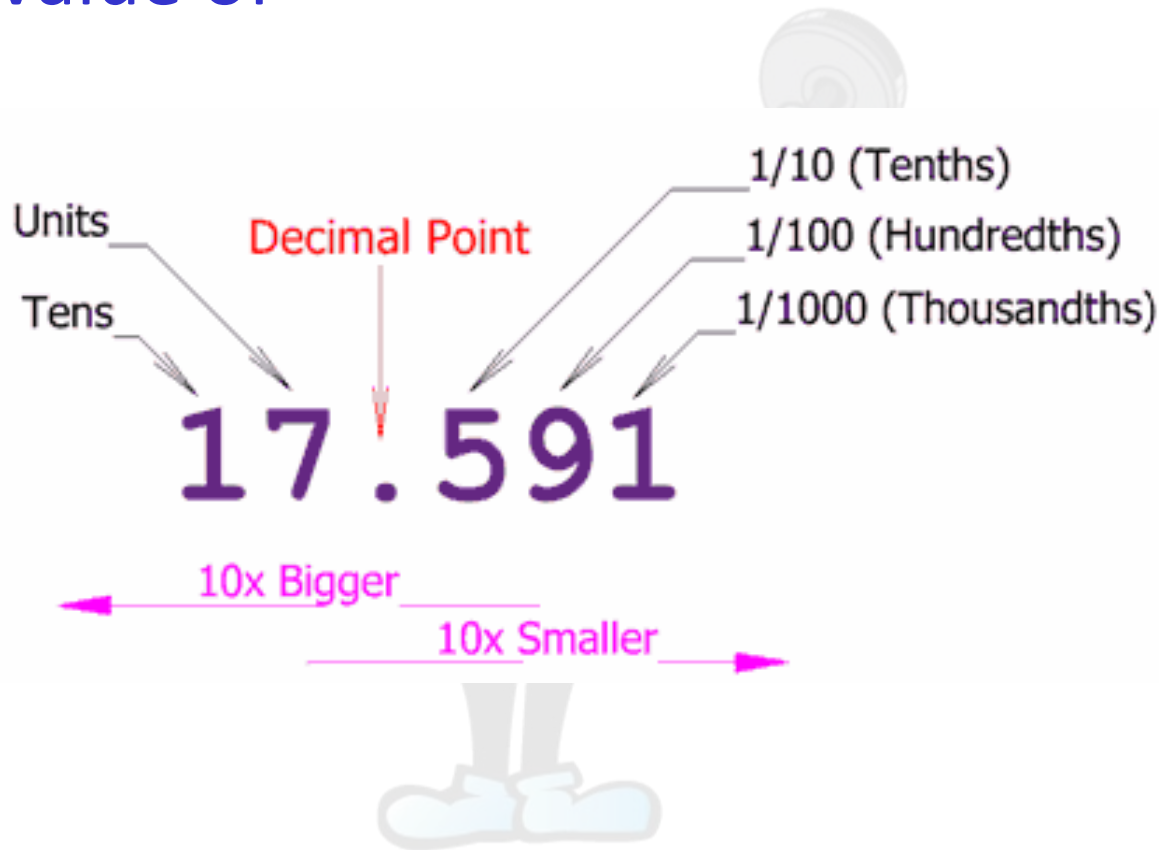
16 and 42

17 and 51

4 and 25

Decimals

- Place value of



Fractions

- Improper fractions to mixed numbers

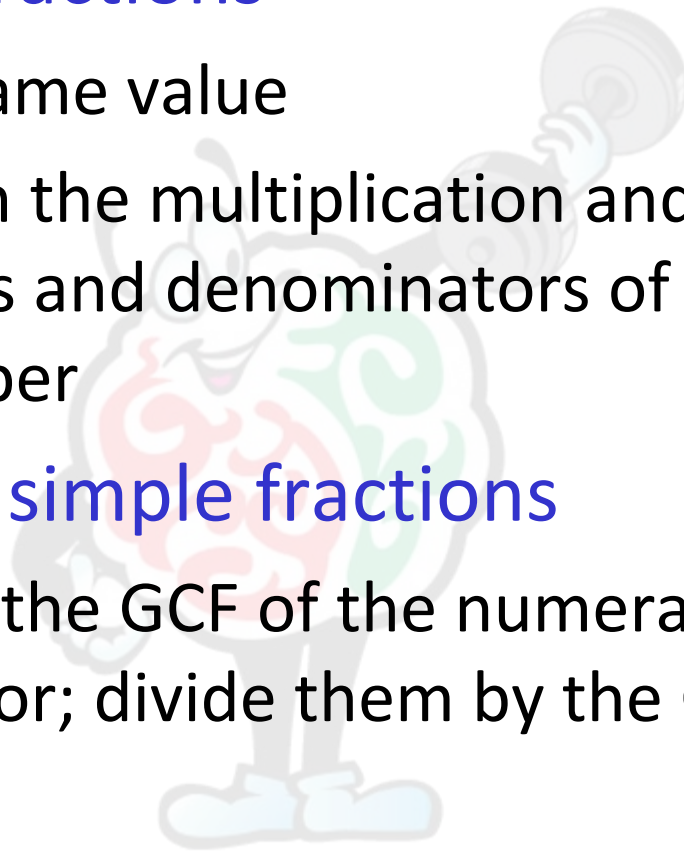
$$\frac{a}{b} = \text{quotient}(a \div b) \frac{\text{remainder}}{b}$$

- Mixed numbers to improper fractions

$$c \frac{a}{b} = \frac{(b * c) + a}{b}$$

Fractions

- Equivalent fractions
 - Have the same value
 - Result from the multiplication and division of the numerators and denominators of a fraction by the same number
- Reducing to simple fractions
 - Determine the GCF of the numerator and denominator; divide them by the GCF



PRACTICE

One-third of one-fifth of 45 is

- A. 5
- B. 24
- C. 75
- D. 3



PRACTICE

Solve:

$$\frac{2}{3} + \frac{5}{7}$$

$$\frac{7}{8} \times \frac{4}{5}$$

$$\frac{3}{4} + 6\frac{5}{4}$$

$$\frac{2}{9} \times \frac{3}{12}$$

$$\frac{30}{40} - \frac{15}{45}$$

$$\frac{4}{5} \div \frac{1}{2}$$

Ratio and proportion

- Ratios may be written as fractions (ratios are also fractions) or by using colons

$$\frac{2}{3} \text{ is the same as } 2 : 3$$

- Proportions are the equality between two ratios (or proportions are equal fractions)

$$\frac{2}{3} = \frac{4}{6} \text{ is the same as } 2 : 3 :: 4 : 6$$

PRACTICE

Simon made a scale model of a car on a scale of 1 to 12.5. The height of the model car is 10cm. What is the height of the real car?

$$3 : 4 :: 30 : \underline{\hspace{2cm}}$$

$$100 : 25 :: \underline{\hspace{2cm}} : 5$$

Find m in the proportion $m/12 = 30/24$.

Sharing/dividing in a ratio

When sharing/dividing any amount/quantity
(e.g., n) in a ratio (e.g., $a : b : c$):

- Add $a + b + c = \text{number of parts}$
- Divide $\frac{n}{\text{number of parts}} = \text{share per part}$
- Multiply $\text{share per part} * a = \text{share of } a$
or b or c or b or c

Rules for ratio and proportion

If $\frac{a}{b} = \frac{c}{d}$, then $ad = bc$.

If $\frac{a}{b} = \frac{c}{d}$, then $\frac{a}{c} = \frac{b}{d}$.

If $\frac{a}{b} = \frac{c}{d}$, then $\frac{b}{a} = \frac{d}{c}$.

Conversion of units

- Know your conversion factors
- milli – centi – deci – BASE UNIT HERE – deka – hecto – kilo
- Or use your mnemonics, e.g.,
King Henry Drinks Much Dark Chocolate Milk
or
Kings Have Diamonds But Diamonds Cost Money

PRACTICE

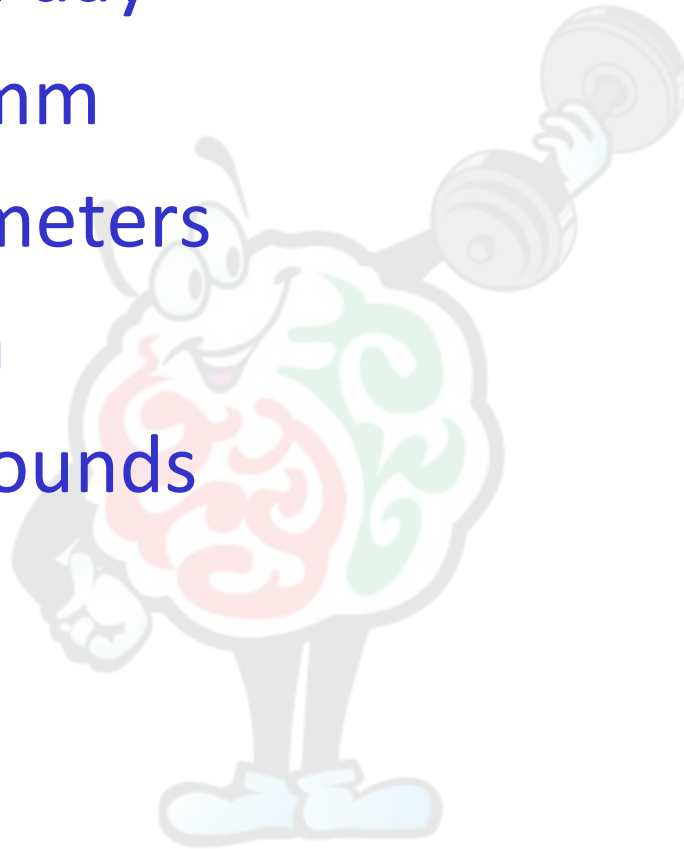
Seconds in a day

3.25 cm to mm

9.24 km to meters

15 km to cm

7.68 kg to pounds



References and further reading

British Broadcasting Corporation. (2012). *GCSE Bitesize: Maths*. Retrieved 27 December 2012, from <http://www.bbc.co.uk/schools/gcsebitesize/maths/>

Fisico, M., L. Sia, J. Ho, Z. Maralit, J. Tan, . . . J. Rosello, & R. Franco. (1998). *21st Century Mathematics Series*. Quezon City: Phoenix Publishing House, Inc.

Pierce, Rod. (2012). *Math Is Fun*. Retrieved 28 December 2012, from <http://www.mathsisfun.com/index.htm>

Stapel, E. (2012). *The Purplemath Forums*. Retrieved 27 December 2012, from <http://www.purplemath.com/index.htm>